UNIVERSITAS INDONESIA

RWA Tokenization with DAO Governance

Blockchain Technology Course

Group 4

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CHAPTER 1: INTRODUCTION

1.1. Introduction

Real World Asset (RWA) tokenization have been a key trend over the last few years. With this system, people can invest their money into a fraction of a RWA, such as real estate. By doing that, they can get benefits such as dividends on every rent to getting a vote on the property management decisions.

One major problem with RWA tokenization is that there have been many cases of financial mismanagement from the third party involved, such as fraud. This leads to many people complaining and court cases that have been done. This mismanagement comes from the centralized approach of the system, which can be solved by leveraging the decentralized system on a blockchain network.

This is where Decentralized Autonomous Organization (DAO) comes in. A DAO based RWA system can help build trust of users and prevent financial mismanagement that could come in a previous system.

Therefore, this project introduces a decentralized RWA tokenization platform governed entirely by a DAO, providing a secure, transparent, and user-centric infrastructure for fractional asset investment.

1.2. Project Aim

The main objective of this project is to develop a platform where the RWA is tokenized, while keeping it safe and trustworthy with DAO. The specific objectives to be achieved include:

- **To design and implement a smart contract infrastructure** capable of tokenizing real-world assets into fractionalized digital tokens safely.
- **To develop a DAO governance mechanism** that allows token holders to vote transparently on critical asset management decisions (e.g., maintenance, dividend distribution, property sale).

- **To build a user-friendly decentralized application (dApp)** interface that enables users to browse, invest in, and manage their fractional RWA portfolios seamlessly.
- **To automate yield distribution** through smart contracts, ensuring that all investors receive their rightful dividends instantly and without third-party interference.

1.3. Problem Scope and Limitations

To maintain focus and ensure the feasibility of the project within the given timeframe, the following boundaries are defined:

- This platform will only support one type of real-world asset, namely real estate properties, as the object of tokenization. Other asset classes such as commodities, securities, or vehicles are not included within the scope of this development.
- Smart contract implementation will be carried out on an Ethereum testnet environment (local Ganache and/or a public testnet such as Sepolia), rather than on the Ethereum mainnet. Consequently, no transactions in this system involve real-valued cryptocurrency.
- The legal validity of token ownership with respect to the underlying physical property is not covered by this project. The platform is a technical prototype that demonstrates the tokenization and DAO mechanisms, and is not a legally recognized investment instrument.
- The process of verifying real-world assets, such as confirming land or building ownership, is performed manually or simulated through an administrator role within the test environment, rather than through integration with an actual third-party verification system.
- The user interface (dApp) is optimized for desktop browser usage only. Mobile responsive support is not a priority in this first iteration.
- The DAO governance mechanism implemented is limited to text-based proposals with token-weighted voting. Advanced governance features such as vote delegation, reputation-based voting, or multi-signature execution are not included in the scope of this project.

1.4. Literature Review

Aloshyna et al. examines how asset tokenization serves as a connecting mechanism between conventional financial systems and the emerging blockchain ecosystem. The authors explore how representing physical assets, such as real estate, commodities, and securities, as digital tokens on a distributed ledger can significantly lower the barrier to entry for investors through fractionalization. The paper highlights that tokenization improves liquidity in traditionally illiquid markets, enables near-instant settlement of transactions, and enhances transparency through immutable on-chain records. It also acknowledges regulatory challenges, particularly around investor protection and the legal recognition of tokenized ownership, which remain key obstacles to mainstream adoption.

Proskurovska et al. takes a more critical stance by examining not only the potential but also the inherent limitations of blockchain-based tokenization when applied at scale to financial markets. The authors argue that while blockchain provides trustless transaction validation, it does not inherently solve off-chain problems such as asset valuation disputes, legal enforceability of token-based ownership, and regulatory compliance across jurisdictions. The paper discusses the tension between decentralization and the need for centralized legal frameworks, pointing out that smart contracts cannot autonomously handle real-world contingencies such as property damage or tenant default. This work is particularly relevant to this project as it underscores the importance of designing robust governance mechanisms, such as DAO-based voting, to handle decisions that cannot be fully automated.

Guo et al. proposes a framework specifically designed to address one of the most pressing technical challenges in RWA tokenization: interoperability across multiple blockchain networks. The authors present xRWA, a cross-chain protocol that allows tokenized real-world assets to be represented and transferred seamlessly between different blockchain ecosystems, such as Ethereum, BNB Chain, and Polygon, without sacrificing the integrity of ownership records. The framework utilizes cross-chain messaging protocols and wrapped token standards to ensure that asset data and transfer events remain consistent across chains. For this project, the xRWA framework serves as a reference point for understanding how future scalability and multi-chain support could be integrated into the platform, even though the current implementation targets a single EVM-compatible network.

Rikken et al. investigate the governance complexities that arise when organizations attempt to operate under DAO structures on blockchain networks. The paper identifies fundamental tensions in DAO design, particularly the trade-off between operational efficiency and fully decentralized decision-making, as large token holder communities often struggle to reach quorum or consensus in a timely manner. The authors also highlight risks such as voter apathy, plutocracy, where wealthy token holders dominate governance outcomes, and the difficulty of modifying smart contract logic once deployed, even when community consensus calls for it. This paper directly informs the governance design of this project, motivating the adoption of proposal thresholds, time-locked voting periods, and proportional voting weights to mitigate these risks while maintaining a fair and democratic governance structure.

Previous literature tends to discuss asset tokenization and DAO governance separately. The novelty of this project is combining both: applying DAO smart contracts to automate management and democratic decision-making for real-world property assets. While existing platforms tokenize assets with centralized management, and while DAO governance has been studied primarily in the context of DeFi protocols, this project integrates both paradigms into a single cohesive system. This enables investors to not only hold fractional ownership of physical assets but also exercise direct, on-chain governance rights over those assets , eliminating the need for trusted intermediaries and reducing the risk of financial mismanagement.

1.3. Competitor Analysis

Table 1. Competitor Analysis

Competitor	What they do well	Their Disadvantages
RealT	Pioneered fractional real estate tokenization on Ethereum; offers consistent rental yield distribution to token holders; has an established user base and a relatively simple onboarding process for U.S. properties.	Governance is fully centralized, RealT management makes all property decisions without token holder input; limited to U.S.-based properties; relies on third-party property managers, creating counterparty risk.
Landshare	Built natively on BNB Chain with lower transaction fees; integrates DeFi features such as staking and yield farming alongside property tokens; active development community.	Governance participation is limited and not fully on-chain; asset selection and management remain under centralized control; smaller asset portfolio compared to more established competitors.

Our Key Values:

- **Full DAO Governance**: Unlike all current competitors, every material decision regarding a tokenized asset, including maintenance approvals, dividend policy adjustments, and property sale, is governed entirely on-chain through transparent token-weighted voting. No centralized party can override the community's decision.
- **Automated and Trustless Dividend Distribution**: Rental income is distributed automatically and proportionally to all token holders via smart contracts immediately upon receipt, removing the need for manual processing or third-party fund managers.

- **Transparent and Auditable Smart Contracts**: All smart contract source code is open-source and verifiable on-chain, allowing any investor or auditor to independently confirm that the system operates as described without hidden administrative privileges.
- **User-Centric dApp Interface**: The platform is designed for accessibility, enabling investors with no technical blockchain knowledge to browse properties, invest, track dividends, and participate in governance through a clean and intuitive interface authenticated solely via their digital wallet.

1.4. Functions/Module

This project will be divided into three main functional modules:

1. Smart Contract Module (Backend Engine):

- Tokenization Function: Responsible for minting property ownership into digital token fractions by setting a maximum supply limit (max_supply). Each token represents a proportional share of the underlying real-world asset, and ownership records are stored immutably on-chain.
- Governance Function (Voting): Allows token holders to submit and vote on proposals (e.g., disbursing funds for a roof repair). Proposals are subject to a defined quorum threshold and a time-locked voting period, after which the outcome is automatically executed by the smart contract without any manual intervention.
- Dividend Distribution Function: Receives rental payments from tenants and distributes them proportionally and automatically to the investors' wallets based on their token holdings at the time of distribution.

2. Frontend Module (User Interface):

- Investor Dashboard: A page for users to view property information, their RWA token balance, accumulated dividends, and transaction history.
- Governance Page: An interface where investors can view active proposals, cast votes, track voting progress in real time, and review the outcomes of past governance decisions.

3. Integration & Identity Module:

- Authentication Function: Secures user entry into the DApp without traditional usernames or passwords, using digital wallets (MetaMask) directly for a seamless and self-sovereign login experience.
- Digital Signature Function: Requires users to consciously approve and sign transactions when transferring funds, purchasing tokens, or casting votes, ensuring that no action is taken without explicit user consent.

1.5. Tools

The project will use the following proposed tools and technologies (with subject to change):

- Ethereum (EVM)
- Solidity
- Truffle & Ganache
- Hardhat
- React
- Node.js
- Web3.js
- MetaMask

PROJECT PLANNING AND SOLUTION DESIGN

2.1. Project Planning

2.1.1 Development Methodology

This project adopts an Agile development methodology using an iteration-based sprint approach. This methodology was chosen because of the multi-layer nature of the project, which involves the integration of blockchain infrastructure, smart contracts, and a user-facing interface, making incremental testing and rapid adjustment essential. Each sprint spans one to two weeks and produces a demonstrable deliverable, ranging from deployable smart contracts to a fully wallet-integrated frontend interface.

2.1.2 Sprint Plan and Timeline

Table 2. Development Sprint Plan

Sprint	**Duration**	**Key Deliverables**
Sprint 1	2 weeks	Smart contract architecture design, Truffle & Ganache setup, deployment of base contracts
Sprint 2	2 weeks	Implementation of DAO voting and dividend distributor contracts, unit testing
Sprint 3	2 weeks	React frontend development, Web3.js and MetaMask integration
Sprint 4	1 week	End-to-end testing, initial security audit, final documentation

The total estimated development period is seven weeks, with one additional week of flexibility as a buffer for re-testing and fixing bugs discovered during the final integration phase.

2.1.3. Risk Management

Table 3. Risk Identification and Mitigation

Risk	Likelihood	Impact	Mitigation
Smart contract bug or exploit	Medium	High	Comprehensive unit testing, code review, use of OpenZeppelin library for battle-tested contract patterns
Voter apathy (quorum not reached)	High	Medium	Adaptive quorum threshold, automated notifications to token holders on active proposals
Frontend-blockchain integration delays	Medium	Medium	Mock API during early sprints, parallel iterative development across layers
Regulatory changes to digital assets	Low	High	Modular contract design so that governance logic can be updated without redeploying the entire system

2.2. Smart Contract Design

2.2.1. Smart Contract Architecture

The system is designed using four modular smart contracts that interact with one another through well-defined interfaces. This modular approach was chosen to enforce separation of concerns and to allow individual components to be updated or replaced without affecting the rest of the system.

Table 4. Smart Contract List and Responsibilities

Contract	Primary Responsibilities
PropertyToken.sol	Minting fractional ownership tokens, managing token transfers, recording investor balances on-chain
PropertyDAO.sol	Accepting governance proposals, managing token-weighted voting, automatically executing approved proposal outcomes
DividendDistributor.sol	Receiving rental income payments, calculating and distributing proportional dividends to all token holders
PropertyRegistry.sol	Registering and verifying asset metadata including location, valuation, and legal document references

CONCLUSION

This project presents a well-planned and technically grounded response to the financial mismanagement risks that are inherent in centralized RWA tokenization platforms currently available on the market. By combining fractional property ownership with a fully on-chain DAO governance system, the platform is designed to eliminate dependence on third-party intermediaries, which have historically been the primary source of risk for investors in this space. The modular smart contract architecture, comprising four interacting contracts, namely PropertyToken, PropertyDAO, DividendDistributor, and PropertyRegistry, provides a solid and extensible technical foundation for both the current implementation and future enhancements.

The structured project plan, organized across four sprints over a seven-week development period, alongside clearly defined team roles and a proactive risk mitigation strategy, provides a strong basis for confidence that the project objectives are achievable within the available time and resources. In the next phase, the deployment and testing of smart contracts on an Ethereum testnet will serve as the primary focus, followed by frontend integration through React to produce a fully functional prototype that can be demonstrated end-to-end.

MidTerm Video: https://youtu.be/Ie1AKFnJpoI

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